An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

1. Executive summary

Floods displace millions of people in India each year. The limiting factor in a rescue is locating survivors, not reaching them.

Three one-metre, 6.4 kg multirotor aircraft search flooded ground as a coordinated group under one operator, identify people from the air, report their position to a metre, and deliver a relief kit to each. No mobile network or internet is required.

Sizing, error budgets and flight software are complete; coordination is demonstrated across three simulated autopilots. No aircraft has flown, so the figures here are calculated.

2. The problem

Boat teams search slowly, at risk, and cannot see rooftops. Helicopters are scarce and committed to evacuation. Conventional drones need one pilot each, depend on infrastructure the flood has removed, and cannot assist a survivor once found.

Detection is the harder problem. A person in floodwater presents head and shoulders, 40 cm across. Detection models take a fixed, small input, so the usual approach shrinks the image and the target with it, to a size at which published accuracy is 29–61 %. A smaller model is worse: smaller models take smaller inputs.

No dataset annotates people in flood imagery. Thermal imaging fails in water, where skin equilibrates toward water temperature.

3. Our solution

System. The operator draws a search region of any shape. The software splits it into three balanced sub-regions and plans a path over each, verified on thirteen non-convex shapes with no path leaving the region and imbalance below 0.02 %. The aircraft fly autonomously; the test harness shows structurally that no command is sent after the mission begins. Detection and geolocation run on board. Each aircraft carries four kits, descends, nulls its groundspeed over a survivor and releases one. Link loss, low battery and operator abort converge on a single return-and-land behaviour.

Detection. The image is not shrunk. Each frame is divided into 48 overlapping tiles at native resolution, and the survey altitude is 40 m rather than 60 m. The target becomes **39 pixels** rather than 13, nine times the area, outside the band where detector performance collapses. Either change alone leaves it inside.

The cost is four times the computation: 48 tiles at 2 Hz is 96 inferences per second, against 130–160 measured for the accelerator. Mission duration is shorter than at 60 m, the lower climb outweighing the extra transects.

Open question. Confirming a survivor across twelve frames needs 3 Hz, which exceeds that capacity. A two-stage filter, discarding the 87 % of tiles that are open water, would recover it. A filter that discards a tile containing a person fails without indication, so it will be evaluated against public data before adoption.

4. Deliverables

1. Three aircraft operating as a coordinated group under one operator.
2. A ground station that displays mission state and is architecturally incapable of commanding the aircraft.
3. Autonomy software — partitioning, path planning, allocation, failsafes — with an automated test suite.
4. Measured detection accuracy and a verdict on the two-stage filter.
5. A publication: tiling, aerial person detection and low-power onboard inference have not been combined for flood conditions.
6. Source code and results, published in either outcome.

5. Benefits

- **Operational.** Ten hectares in under eight minutes, returning coordinates rather than a video feed needing continuous attention.
- **Economic.** The complete system costs less than one hour of helicopter search time, travels by road, and is operated by two people.
- **Institutional.** A thrust stand, ground station, tested codebase and trained students remain with the department.
- **Strategic.** Indian suppliers wherever they meet the specification; domestic lead times are half those of imports.

6. Resources needed

Funding is requested in 32 small steps rather than as one grant. Each step buys one named set of parts and is released only once the previous step has produced a stated result. If a step fails, the programme stops there and nothing beyond it has been spent.

The order follows engineering dependency: one motor is measured before the other eleven are bought, and the first aircraft flies a complete mission before the second is begun.

Phase schedule

The third column is the amount released. The fourth is what must already be true before it is released.

#	Objective	INR	Released only after
1	Establish thrust and mass measurement capability	26,196	Approval
2	Verify static thrust against the 3.18 kgf requirement	5,175	Instruments commissioned
3	Aircraft 1 — autopilot and control link	25,990	**Thrust verified**
4	Aircraft 1 — onboard computer and AI accelerator	21,850	Autopilot bench-tested
5	Aircraft 1 — centimetre positioning (RTK receiver)	27,892	Compute stack operating
6	Aircraft 1 — camera and radios	29,440	Position fix acquired
7	Aircraft 1 — airframe fabrication	20,144	Avionics integrated
8	Aircraft 1 — battery pack assembly	29,299	Airframe fabricated
9	Aircraft 1 — speed controllers, propellers, release servos	27,098	Pack bench-discharged
10	Aircraft 1 — propulsion completed	15,525	Drive train installed
11	Charging and manual override for flight test	25,300	Aircraft assembled and weighed
12	Field equipment: consumables, sun hood	17,274	Charging verified
13	Aircraft 2 — autopilot and control link	25,990	**Aircraft 1 flew a full mission**
14	Aircraft 2 — onboard computer and AI accelerator	21,850	Autopilot bench-tested
15	Aircraft 2 — centimetre positioning (RTK receiver)	27,892	Compute stack operating
16	Aircraft 2 — camera and radios	29,440	Position fix acquired

#	Objective	INR	Released only after
17	Aircraft 2 — airframe fabrication	20,144	Avionics integrated
18	Aircraft 2 — battery pack assembly	29,299	Airframe fabricated
19	Aircraft 2 — speed controllers, propellers, release servos	27,098	Pack bench-discharged
20	Aircraft 2 — propulsion completed	20,700	Drive train installed
21	Aircraft 3 — autopilot and control link	25,990	**Two aircraft, separation verified**
22	Aircraft 3 — onboard computer and AI accelerator	21,850	Autopilot bench-tested
23	Aircraft 3 — centimetre positioning (RTK receiver)	27,892	Compute stack operating
24	Aircraft 3 — camera and radios	29,440	Position fix acquired
25	Aircraft 3 — airframe fabrication	20,144	Avionics integrated
26	Aircraft 3 — battery pack assembly	29,299	Airframe fabricated
27	Aircraft 3 — speed controllers, propellers, release servos	27,098	Pack bench-discharged
28	Aircraft 3 — propulsion completed	20,700	Drive train installed
29	Ground base receiver, the source of the corrections	27,892	First flight completed
30	Stable, repeatable base mounting	4,600	Base receiver acquired
31	Deployable ground segment	28,750	Base station operating
32	Statutory registration	5,750	Ground segment complete
	Total	**7,43,004**	

Phase 3 commits the first significant expenditure and is released only on a measured thrust figure. Phase 13 begins the second aircraft only after the first has flown a complete mission, so the largest blocks of expenditure fund replication of a demonstrated configuration rather than a projection.

Component selection

The schedule gives the amount *released* per phase. The tables below give the parts each phase *buys*. The two differ by a fixed rule: a released amount is the parts cost plus 22 % customs duty on the imported share, plus 18 % GST, plus 15 % contingency. Across the programme that is INR 5,95,168 of parts against INR 7,43,004 released, a factor of 1.25.

Every line in the tables below is now a live listing from a named supplier, priced and linked. An earlier revision of this brief said the opposite — that seven lines were firm and that "every other line in this section is a quotation or an estimate from a comparable part." That is no longer true of any line. The two that mattered most have both settled: the GNSS receiver, previously an INR 18,000 estimate against an unpriced part, is quoted by the supplier at INR 25,000 with RTK rover operation confirmed; and the flight controller, camera and speed controllers, all previously carried at quotation, are ordinary retail. The tables are generated from that basket rather than transcribed, so they cannot drift from it.

Three consequences worth stating plainly, because they move the figure in both directions. The onboard computer was budgeted at INR 8,000 against a real price of INR 19,999, which is the single largest correction in the programme. The certified scale, budgeted at INR 12,000, is not in the ask at all: the department already holds one, so it is an institutional contribution rather than a purchase. And the parts that were to buy a 5.8 GHz mesh now buy three separate radios, because the mesh was withdrawn on its link budget.

Terms used below. *RTK*: satellite positioning corrected against a second receiver on a surveyed point, giving centimetre rather than metre accuracy. *6S3P*: eighteen cells, six in series for voltage and three in parallel for capacity. *kgf*: thrust written as the weight it would lift. *TOPS*: trillion operations per second, the accelerator's throughput. *ESC*: the electronic speed controller between battery and motor. *BMS*: the board that balances and protects the cells — there is none in this build, and the note after the tables says why.

Measurement instruments — phases 1–2 INR 3,135 of parts

Item	Model	INR, all 3	Why this part, and from whom
Load-cell amplifier	7Semi HX711 24-bit ADC	104	*Robu*
Load cell	REES52 20 kg load cell + HX711 breakout	854	The motors ship without a thrust curve. Measuring thrust and current together gives thrust per watt, which is the number the whole sizing loop rests on. Commercial stands start near INR 45,000; this is the same measurement for INR 1,331 of parts and a printed frame. *Amazon India*
Thrust-stand mast	SURYAKUSH metal 2.1 m tripod	373	*Flipkart*
Fasteners	Hex socket-head assortment M3/M4/M5, SS304	1,140	*OnlyScrews*
Threadlocker	Loctite 243, 50 ml	664	*Amazon India*

Per aircraft, avionics and sensing — phases 3–6, repeated at 13–16 and 21–24 INR 81,399 per aircraft, INR 3,44,197 in all

Item	Model	INR, all 3	Why this part, and from whom
Flight controller	Holybro Pixhawk 6C Mini, 3 off	66,147	Dual IMU, Pixhawk standard, ArduPilot native. Was carried at quotation; now a live retail listing. *Robu*
FC vibration mount	Glass-fibre anti-vibration shock absorber, 3 off	483	*Robu*
Companion computer	Raspberry Pi 5, 8 GB, 3 off	59,997	The host the autonomy runs on — Linux, ROS 2, the coverage planner, MAVLink routing and the delivery logic. The accelerator below is an M.2 card over PCIe and the camera is a CSI module, so neither has anything to plug into without it. *Robu*
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS, 3 off	35,247	Hailo-8, 26 TOPS. The requirement is 96 inferences/s at 640 px (48 tiles at 2 Hz), which this clears; 26 TOPS is margin on the one budget that already fails, not a specification floor. *Robu*
Compute cooling	Official Raspberry Pi 5 active cooler, 3 off	1,557	*Robu*
Storage	SanDisk High Endurance 128 GB microSD, 3 off	11,967	*Flipkart*
GNSS RTK receiver	Teravolt AeroNav-Pro RTK, 3 rover + 1 base, 4 off	1,00,000	Four units: one rover per aircraft and one base. Centimetric RELATIVE geometry is what lets recall and delivery error be measured against surveyed ground. Supplier has confirmed RTK rover operation and quoted; this line was an INR 18,000 estimate and is now firm. *Teravolt*
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens, 3 off	20,397	12.3 MP at 1/2.3 in and a 6 mm lens give 1.03 cm/px at 40 m, which puts a person in water at 39 px — just over the COCO small-object threshold. Was carried at quotation. *Robu*
Video transmitter	SunRobotics TS832, 5.8 GHz, 600 mW, 3 off	12,717	Analog, one channel per aircraft. Removes video from the data network and, more importantly, removes three software H.264 encodes from the board that runs tiled inference. *IndiaMART*

Item	Model	INR, all 3	Why this part, and from whom
Command receiver	ExpressLRS RX24T, 2.4 GHz, 3 off	4,317	The only path that can command the aircraft, reaching the flight controller directly rather than through the autonomy stack. Must run firmware 3.5+ in native MAVLink mode, which carries control and telemetry on one link and removes a second radio. *Robu*
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm, 3 off	2,967	SX1262 in the 865–867 MHz delicensed band. Carries mission data and swarm coordination at 54 dB of margin — the largest in the system, which is where the survivor coordinates belong. *Robu*
Power module	Holybro PM07, 3 off	15,747	*Robu*
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in, 3 off	11,817	*Amazon India*
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC, 3 off	837	*Robu*

Per aircraft, airframe and drive — phases 7–10, repeated at 17–20 and 25–28 INR 58,271 per aircraft, INR 1,89,475 in all

Item	Model	INR, all 3	Why this part, and from whom
Motors	Tarot TL96020, 5008, 340 KV, 12 off	56,436	3.18 kgf each, giving twice the aircraft weight in thrust; 340 KV suits an 18 in propeller at this pack voltage. Twelve, of which the first is measured on the stand before the remaining eleven are committed. *Robokits*
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair, 16 off	26,256	Sixteen for twelve positions: the most frequently replaced item in a flight-test programme, and the four spares are the whole spares policy. *Robokits*
Speed controllers	Darkmatter VISHNU 50 MK-III, 4-in-1, 50 A, 3 off	15,567	One 4-in-1 board drives all four motors. 50 A continuous against a 29 A per-motor peak. Made in India, and was carried at quotation. *Robu*
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm, 6 off	13,554	25×23 mm carbon. Bending is not the driver at a safety factor of 24; the clamp and joint design is. *Robu*
Motor mounts	Tarot 25 mm motor mount, TL9602, 12 off	13,428	*Robu*
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs, 3 off	9,036	*Amazon India*
Landing gear	F450/F550 landing skid, 3 off	1,647	*Robu*
Printed parts	Pro-Range PETG-CF filament, 1.75 mm, 1 kg	949	*Robu*
Release servos	MG90S mini servo, 180 deg, 12 off	1,788	*Robocraze*
Cells	Molicel INR21700-P45B, 6S3P, 54 off	27,486	Molicel P45B, 6S3P, 292 Wh. Covers the mission, a full second search and four minutes of holding within an 80 % depth of discharge. *Robu*
Cell holders	3x1 21700 holder, 21.75 mm bore, 54 off	648	*Robu*
Pack interconnect	Pure nickel strip, 0.15 x 27 mm	6,500	*IndiaMART*

Item	Model	INR, all 3	Why this part, and from whom
Group interconnect	Tinned copper braided jumper	500	*IndiaMART*
Balance leads	JST-XH 6S, 200 mm, 22 AWG, 3 off	372	*zBotic*
Pack fusing	ANL fuse holder + 150 A ANL fuses, 3 off	9,000	150 A ANL fuse per pack. Short-circuit protection that cannot nuisance-trip at the 115 A thrust peak. *Njour*
Pack retention	300 mm LiPo strap, reusable, 6 off	372	*Robu*
Main leads	10 AWG ultra-flexible silicone wire	149	*Robu*
Power connectors	Amass XT90S anti-spark, M/F pair, 12 off	2,160	*Robokits*
Signal connectors	JST ZH 6-pin, 200 mm leads, 3 off	69	*Robu*
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters, 3 off	3,417	*Desertcart*
Insulation	PVC heat-shrink sleeve, 150 mm, 3 off	141	*Robu*

Flight-test support — phases 11–12 INR 38,925 of parts

Item	Model	INR, all 3	Why this part, and from whom
Safety-pilot transmitter	RadioMaster TX12 MKII ELRS + RP1 receiver	16,519	ExpressLRS, EdgeTX, with a spare receiver. Manual override independent of the autonomy stack, which is a rule requirement and a safety one. *zBotic*
Battery charger	ToolKitRC M6D, 500 W, 25 A, dual	7,349	500 W dual-channel. Two 292 Wh packs recharged between sorties is the constraint on flight-test throughput. *Robu*
Pack health monitor	Holybro PM08-CAN, 14S, 200 A	9,058	200 A CAN power module on the bench, to log pack internal resistance as the cells age. The sag simulation currently assumes a DC-IR; this measures it. *Robu*
Cable management	Nylon zip ties, 350 mm, 100 pcs	135	*Robu*
Heat-shrink kit	328 pcs assorted 2:1 heat-shrink	159	*Robu*
Mounting tape	3M VHB 5952, 6 mm x 8.2 m	560	*Amazon India*
Hook and loop	Self-adhesive hook and loop, 25 mm x 2 m	145	*Amazon India*
Consumables	Fasteners, tape, connectors, filament	5,000	*in-house*

Ground segment and statutory — phases 29–32 INR 19,436 of parts

Item	Model	INR, all 3	Why this part, and from whom
Video receivers	RC832S 5.8 GHz AV receiver, 3 off	13,197	Three, one per aircraft feed. The rules require the ground station to display a live camera feed from each aircraft simultaneously. *Robu*
Receive antennas	Triple-feed 5.8 GHz patch, SMA, 3 off	2,517	12 dBi patch at the ground station, which is where the 28.5 dB of video margin at the 600 m geofence comes from. *Robu*
Video capture	USB 2.0 analog AV capture adapter, 3 off	1,197	*Robu*
Coordination base	EByte E22-900T22U, SX1262, USB	1,646	The ground end of the 865 MHz link, over USB to the ground station. *Hubtronics*
Base station mount	Photographic tripod and adapter	879	*Amazon India*

Why there is no battery management board. An earlier revision funded "6S 60 A BMS, distribution board, regulator" as one INR 8,500 line. Two thirds of that is bought above under other names: the distribution board is the Holybro PM07 and the regulators are the two UBECs. What remained was cell protection, and a board in series with the pack is the wrong device for it at this current.

The pack draws 42 A in hover and 115 A at the thrust peak. A 60 A board trips in normal flight; the 100 A board stocked in India trips at full thrust, and a cut-out at maximum thrust is a crash rather than a protection. A board rated past 115 A would put its transistors in series with the main path, adding resistance and heat exactly where the pack already reaches its low-voltage floor. So the protection is distributed across parts that are in the tables: a 150 A fuse for short circuits, the flight controller's low-voltage failsafe for over-discharge, balance leads charged on the bench, and the PM07 for current and voltage telemetry. Per-cell voltage is read on the charger, which sees all six cells through the balance lead on every charge — which is when a diverging group shows up. What this build does not have is per-cell protection *in flight*, and that is a stated consequence of not fitting a series board rather than an oversight.

What this arrangement costs us

Thirty-two approvals is a material administrative overhead, and buying one aircraft at a time places the same imported order three times — roughly eight to twelve weeks of schedule. Consolidating the per-aircraft phases into one order is available if schedule becomes binding.

The detection evaluation needs no procurement: about a week on a departmental GPU and 30 GB of storage, against a public dataset.

7. Future work

Near term. Measure what is currently calculated: detection accuracy, motor thrust, setup time. Evaluate near-infrared sensing — water absorbs strongly and skin does not, so a person appears bright against dark water. A camera without its infrared filter costs INR 3,000, and a public dataset allows evaluation before purchase. Only the training data is flood-specific; the same system suits earthquake, wildfire and avalanche search.

Longer term. A capable aircraft deciding for itself within a small power budget is a hard requirement in planetary aviation. *Ingenuity* flew 72 times on Mars from a 1.8 kg airframe in an atmosphere under one per cent of Earth's density; *Dragonfly* launches in 2028 for Titan. A command takes minutes to arrive, so every decision is made on board. What transfers is what is not flood-specific: onboard survey planning, perception run to an energy budget, and safe behaviour without a link. Microcontroller-class processors, often proposed for such missions, cannot resolve targets of this size, so a planetary configuration needs a different operating point.

8. Conclusion

Three aircraft, one operator, no network: survivors located in flooded terrain and reached with a relief kit. Eight minutes for ten hectares, returning coordinates a rescue coordinator can act on. The design is specified; what remains is to build it and measure it.

Exposure is never more than one phase. The first is INR 31,371 and settles whether these motors lift this aircraft; if they do not, the programme stops there and the department keeps a thrust stand. Every later release is backed by the previous result, and the two largest come only after a complete aircraft has flown.

Floods recur every monsoon, and no capability of this kind exists at a scale two people can deploy. The whole system costs less than an hour of helicopter search time. We ask the department to release Phase 1.